

USB-C DAUGHTERBOARD / ENGINEERING DRAFT

Upstream: USB-C, USB 3.2 Gen 2 data UFP, USB-PD power source, 20 V / 5 A maximum.
Downstream: 2 Type-C + 3 Type-A SuperSpeed, 1 Type-A USB 2.0.
Use two GPU power inputs OR one EPS input. Not for simultaneous independent supplies.

Power Supplies and Inputs

File: Power.kicad_sch

MCU and Upstream PD Control

File: MCU.kicad_sch

USB Hub and Downstream Ports

File: USB_Hub.kicad_sch

12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
Engineering draft – not released for manufacture

Sheet: /
File: XG_Mobile_USB_Hub.kicad_sch
Title: XG Mobile USB Hub

Size: A3	Date:	Rev: DRAFT A
KiCad E.D.A. 9.0.9		Id: 1/4

The diagram is a schematic representation of a USB-C daughterboard, organized into three main functional blocks arranged vertically. The background features a light gray grid with horizontal lines labeled A, B, C, D, E, and F, and vertical lines labeled 1 through 8.

Block 1: Power Supplies and Inputs

- Located in the upper section, spanning from horizontal line A to B.
- Contains a large, empty rectangular box representing the power supply and input circuitry.
- Below the box, the text "File: Power.kicad_sch" is displayed.

Block 2: MCU and Upstream PD Control

- Located in the middle section, spanning from horizontal line C to D.
- Contains a large, empty rectangular box representing the microcontroller and upstream power delivery control circuitry.
- Below the box, the text "File: MCU.kicad_sch" is displayed.

Block 3: USB Hub and Downstream Ports

- Located in the lower section, spanning from horizontal line E to F.
- Contains a large, empty rectangular box representing the USB hub and downstream port circuitry.
- Below the box, the text "File: USB_Hub.kicad_sch" is displayed.

Title Block (Bottom Right):

- A rectangular box containing project information.
- Text inside the box: "12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream", "Engineering draft – not released for manufacture", "Sheet: /", "File: XG_Mobile_USB_Hub.kicad_sch", and "Title: XG Mobile USB Hub".
- Below the title, a table provides additional details:

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The diagram is a schematic representation of a USB-C daughterboard, organized into three main functional sections arranged vertically:

- Power Supplies and Inputs**: The top section, containing a large rectangular area representing the power management circuitry.
- MCU and Upstream PD Control**: The middle section, containing a large rectangular area representing the microcontroller and upstream power delivery control logic.
- USB Hub and Downstream Ports**: The bottom section, containing a large rectangular area representing the USB hub and downstream port connections.

At the bottom right corner, there is a detailed information box:

- Specifications:** 12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
- Status:** Engineering draft – not released for manufacture
- Sheet Information:** Sheet: / File: XG_Mobile_USB_Hub.kicad_sch
- Title:** XG Mobile USB Hub
- Metadata Table:**

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The image displays a schematic diagram for a USB-C daughterboard, presented as an engineering draft. The title at the top left reads "USB-C DAUGHTERBOARD / ENGINEERING DRAFT". Below the title, there are three main functional blocks arranged vertically:

- Power Supplies and Inputs**: This block contains a large rectangular area representing the power supply circuitry.
- MCU and Upstream PD Control**: This block contains a large rectangular area representing the microcontroller unit and upstream power delivery control logic.
- USB Hub and Downstream Ports**: This block contains a large rectangular area representing the USB hub and downstream ports.

At the bottom right corner, there is a detailed information box containing the following text:

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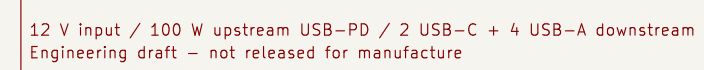
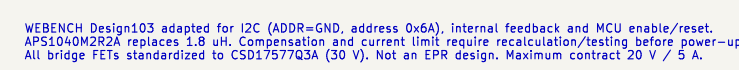
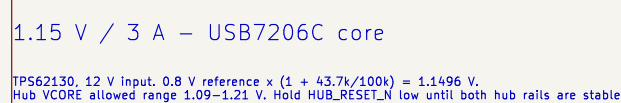
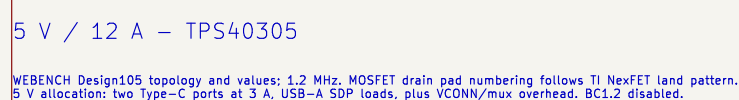
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Named nets connect the circuit blocks on this sheet and across the hierarchy

Choose GPU pair OR EPS with a break-before-make selector rated $\geq 30\text{ A}$, $\geq 24\text{ VDC}$. Do not switch under load.
GPU pair MUST come from the same PSU. Both GPU connectors required for full output. Do not combine separate PSUs.
Connector pin numbering is the electrical interface: verify keying, mating face and selected manufacturer drawing before layout.



Sheet: /Power Supplies and Input Files: Downloaded on:

Title: Power Supplies and Input

Size: User	Date:
KiCad E.D.A. 9.0.9	

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MCU and Upstream PD Control

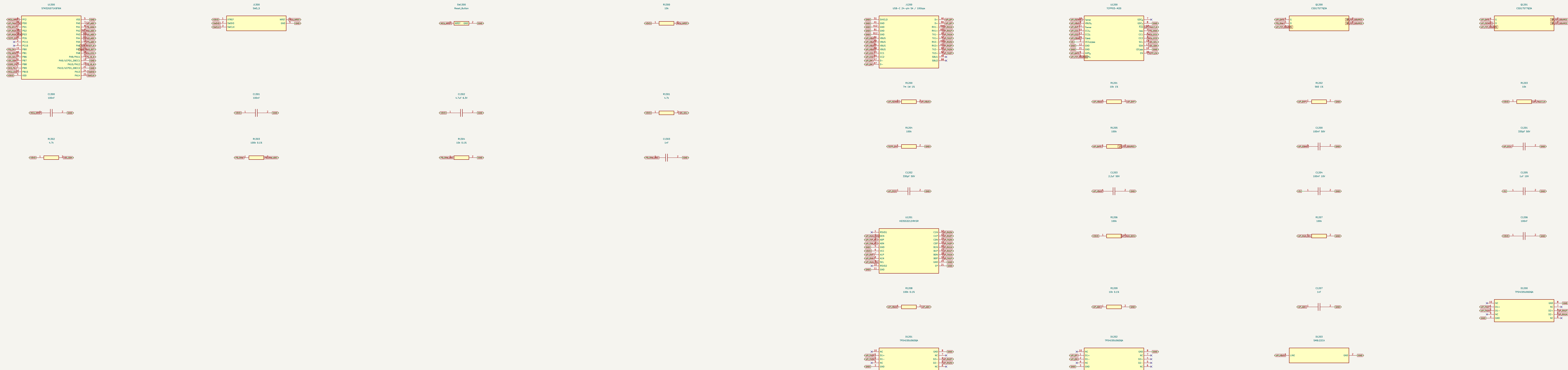
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STM32G071 – USB-PD policy and power control

STM32G071KBT6N LQFP32: PA8=UCPD1_CC1, PB15=UCPD1_CC2, PB6/PB7=I2C1, PA13/PA14=SWD.
Source-only power path. UCPD dead-battery pins grounded; disable dead-battery function in firmware.
Boot from flash using option bytes; preserve NRST function. Firmware required; no PD stack is generated here.

Upstream Type-C – USB data UFP / PD source

STM32 UCPD negotiates PD and manages data-role swap to UFP while keeping power-source role.
TCPD03 source path: common-source back-to-back FETs. Sink path not fitted, 7 mOhm gives nominal 6 A hardware OCP.
100 W requires identified 5 A e-marked cable; otherwise cap contract to 3 A. Hardware OVP ceiling -22 V.

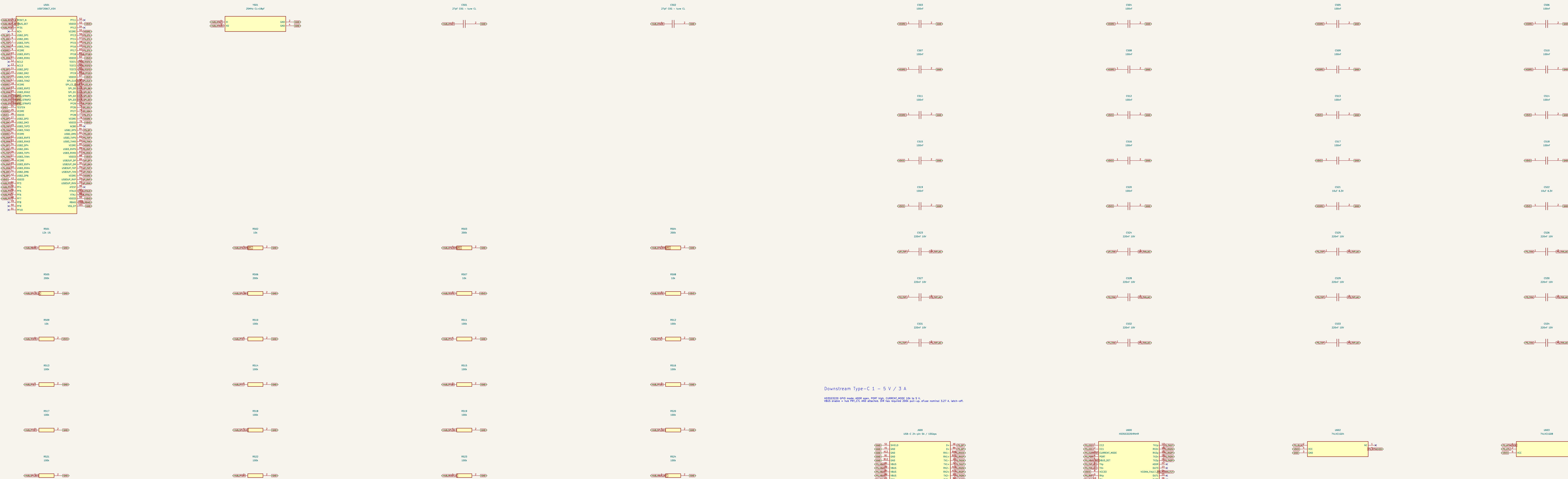


USB Hub and Downstream Ports

Named rats connect the circuit blocks on this sheet and across the hierarchy.

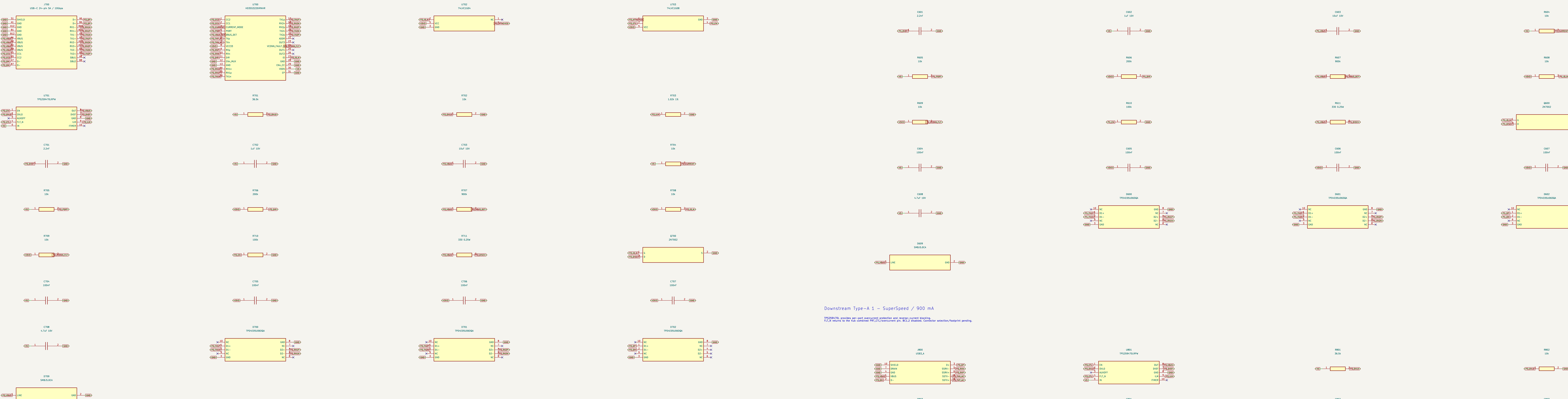
USB7206C – six downstream ports

Part 1/2: Type-C, Parts 3/4/6: Type-d, SuperCoated, Part 8: Type-d USB 2.0.
Configuration 3 per 0120003880? Tactile 3-1/4"/1-5, RESET and IRUS, DAT driven by PS MCU, not high-voltage VSGT.



Downstream Type-C 2 – 5 V / 3 A

W99553229 GPIO mode, W998 open, PORT high, CURRENT MODE 50A to 5 V,
FRTS enable = bus FRTS_C, and unswitched. JIR has required 200A pull-up, if use nominal 5.27 A, latch-off.



Downstream Type-A 2 – SuperSpeed / 900 mA

TP1259470, provides per-port overcurrent protection and reverse-current blocking.
FLT_N returns to the bus combined PFLT_N/overcurrent pin. BCL2 disabled. Connector selection/footprint pending.

